Explanation: Why Two Wager Tiers Exist

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This page explains the idea behind Safe Bet and Smart Bet, in plain language. For the full math, worked proofs, and citations, read `docs/white-paper.md`.

Two different questions

A wager decision can answer two different questions. This project treats them as separate tools, not one blended answer.

Question 1: "What wager can't hurt me if I'm right?"

This question ignores confidence entirely. It asks: assuming every opponent plays their single most threatening legal move — betting everything and answering correctly — what wager still keeps me safe? This project calls the answer **Tier 1**, shown in the app as **Safe Bet**.

Question 2: "Given how sure I actually am, what wager gives me the best real chance to win?"

This question needs two more pieces of information: your own confidence in the category, and some sense of how opponents tend to wager and answer. This project calls the answer **Tier 2** for Final Jeopardy, and the **Daily Double engine** for a mid-game Daily Double. The app shows this as **Smart Bet**.

Why Safe Bet can recommend more money than Smart Bet

Safe Bet has to plan for the worst realistic case, every time. Smart Bet gets to use your actual confidence, and it counts a tie for first as a win — matching the real show rule that co-champions both advance to the next game. A wager that only ties an opponent's worst case can be exactly as good, in real win probability, as a much larger wager that beats it outright.

Why the Daily Double engine gives a range, not one number

A Final Jeopardy wager only affects the next thirty seconds. A Daily Double wager affects the rest of the game — more clues, maybe another Daily Double, then Final Jeopardy after all of that. There is no clean, exact formula for that. Instead, the engine simulates the rest of the game forward, thousands of times, for each candidate wager, and counts how often each one wins.

Every simulated result carries some uncertainty, the same way a poll of 1,000 people carries a margin of error. When several wagers are close enough that the simulation cannot tell them apart, the app shows this honestly as a shaded **plateau** on the chart, instead of picking one number that looks precise but is not. The app always recommends the smallest wager in that plateau — there's no reason to risk more money than the evidence supports.

What is proven, and what is a placeholder

The Safe Bet math is fully determined. It needs no assumption about anyone's odds of being correct, so there is nothing to calibrate.

Smart Bet and the Daily Double engine both need a guess at how opponents behave — how often they get a clue right, and how they tend to size their own wagers. This project uses clearly labeled placeholder numbers for that guess, not numbers fitted to real historical games. The *shape* of the advice (do not overbet past the point of no added benefit; prefer smaller wagers among ties) holds regardless. The *exact* win-probability numbers should not be quoted as validated statistics yet. See `docs/adr/0002-opponent-model-assumptions.md` and `RISK_REGISTER.md`.

Related pages

- `docs/white-paper.md` — full math, proofs, and citations.
- `docs/tutorial/getting-started.md` — see both tiers in action with known numbers.
- `docs/reference/engine-api.md` — the exact functions behind these ideas.